Daniel Drake

Tarentum, PA
ddrakepgh@gmail.com | 724-493-2169

Summary

Software Engineer with extensive experience in distributed storage systems, networking, robotics, and scientific instrumentation. Proven background designing resilient, high-availability systems for enterprise storage platforms and automated imaging systems used in drug discovery. Experienced across startups and large-scale enterprise environments, with strengths in C++, Linux systems programming, hardware/software integration, robotics control systems, and technical leadership.

Technical Skills

Languages: C++ (various dialects), C, Python, Go (Every shipping product written in C++)
Systems & Domains: Distributed systems, clustered storage, networking, instrumentation control, Linux systems programming, image acquisition and processing
Tools: Git/GitHub, Linux, GCC/Clang, GDB, VSCode, Claude-Code, Jira, Confluence
Practices: Performance optimization, engineering for resiliency and serviceability, Agile/Scrum, technical leadership

---

Professional Experience

NetApp — Senior Engineer / Customer Support / Engineering Manager — 2004 - present

- Developed resiliency and automated recovery capabilities for distributed storage networking systems supporting enterprise and hyperscaler deployments.
- Focused on intra-cluster network reliability, fault detection, and automatic remediation
- Supported early-adopter customers by owning issue triage, defect resolution, communication, and finding workarounds to stabilize clusters..
- Led Windows protocol (CIFS/SMB) integration efforts for clustered storage appliances.
- Managed engineering schedules, feature delivery, cross-functional teams.

Spinnaker Networks — Startup — 2000-2004 — Software Engineer

- Assembled a team to implement CIFS/SMB network protocol support for a high-performance clustered file server running on custom hardware.
- Led clean-room reverse engineering of CIFS/SMB networking protocols.
- Company acquired by NetApp

Cellomics — 1997 - 2000 — Lead Software Engineer

- New company created to productize prior company's instruments.
- Focused on resiliency while improving fault handling and supportability.
- Improved storage and acquisition pipelines for high-throughput image processing.
- Company acquired by Fisher Scientific.

Biological Detection Inc. — CMU Spinoff — 1994-1997 — Programmer

- Developed control software for automated drug discovery instrumentation using robotic motion systems and confocal microscopy.
- Implemented robotic arm control for 96-well plate handling and micro-pipette dispensing systems.
- Wrote software for microscope control, image acquisition, 3-axis stage positioning, and image processing.

Biological Detection Systems — CMU Startup — 1990-1994 - Programmer

- Created multi-wavelength biological imaging systems using fluorescence microscopy techniques labeling cellular and subcellular structures.
- Implemented support for 3-axis stage positioning, wavelength filters, and image acquisition hardware using Peltier-cooled CCD cameras.
- Supported custom hardware integration and image-processing workflows.

Robotics & Embedded Systems Projects

FIRST Robotics Competition Mentor

- Advised high school robotics teams on motion control, PID systems, sensor integration, and computer vision.
- Worked with accelerometers, gyroscopes, Hall effect sensors, encoded stepper motors, and vision-guided navigation systems.

6502 Breadboard Computer

- Built a custom 6502-based breadboard computer including ROM, RAM, address/data buses, 6522 VIA, and LCD display.
- Experimenting with emulating memory-mapped I/O using Raspberry Pi Pico PIO state machines.

Arduino-Based Control Systems

- Reverse engineered and replaced remote control electronics for an out of support cast iron stove using Arduino and relay control hardware.
- Requirements were simplicity, reliability and fault-tolerant operation

Jetson Nano Robotic Car

- Built a Jetson Nano-based mobile robot cat with camera streaming and web-based remote control.
- Working on autonomous navigation and image-recognition capabilities.

Education

American University — Washington, D.C.
B.S. with minor in Computer Science
Additional graduate-level coursework in Computer Science